

Aesthetic Breast Surgery

The Study of the Safety and Effectiveness of Motiva SmoothSilk Silicone Gel-filled Breast Implants in Patients Undergoing Primary and Revisional Breast Augmentation: 5-Year Clinical Data

Caroline Glicksman, MD, MSJ[®]; Andrew Wolfe, MD;
and Patricia McGuire, MD[®]

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Abstract

Background: The FDA requires implant manufacturers to continue collecting safety and efficacy data annually through Year 10. Investigators are encouraged to follow their enrolled patients in person to obtain the most accurate and complete data.

Objectives: The authors reviewed the 5-year data on the safety and effectiveness of Motiva SmoothSilk silicone gel-filled breast implants submitted to the FDA. The reported data include both the primary and revisional breast augmentation cohorts.

Methods: The 5-year clinical data from the breast augmentation cohorts of the Motiva IDE 10-year pivotal study were submitted to the FDA in 2024. Data were collected on adverse events, reoperation, patient and physician satisfaction, connective tissue and rheumatologic diseases, and quality-of-life instruments. MRIs were obtained at Years 1, 2, 3, and 5 for patients enrolled in the MRI substudy.

Results: There were 451 primary augmentation patients and 109 revisional augmentation patients enrolled in the pivotal study. Follow-up rates were 87% of expected patients at both Years 4 and 5. Of the 218 patients enrolled in the MRI cohort, 152 underwent MRI screening at Year 5, an overall compliance of 79.2% of expected screenings. The reported reoperation rates for any reason were 8.8% in the primary augmentation cohort and 36.0% in the revisional augmentation cohort.

Conclusions: The 5-year data from the primary and revisional breast augmentation cohorts reveal low rates of device rupture. Size change and malposition were the most common causes of reoperation in the primary augmentation cohort, and capsular contracture and size change were the leading drivers of reoperation in the revisional augmentation cohort.

Level of Evidence: 2 (Therapeutic)

Breast implants were first introduced in the 1960s. Since that time, advances in materials science, chemical engineering, and implant biology have led to significant improvements in implants and their use. Establishment Labs, the maker of Motiva (Establishment Labs Holdings, Inc., Alajuela, Costa Rica) implants, is 1 of more than 13 manufacturers of silicone breast implants worldwide. Motiva SmoothSilk implants have been available outside of the United States since 2010 and have undergone updates in shell structure and gel technology since then. Much of the pre-FDA trial literature on the safety and efficacy of Motiva implants was based on the real-world experience of the initial adopters of the new implant technology. These early papers, documenting the initial experiences of surgeons in Korea, Brazil, Italy, Germany, and the UK, were retrospective, single-arm studies or had small numbers of patients included.^{1,2,3,4} In 2021, Doloff et al published their preclinical animal research on the

topography of the Motiva SmoothSilk surface. They demonstrated the potential benefits of the 4-micron surface's reduced inflammation and foreign body response.⁵

The US pivotal study began enrollment in 2018 and included 2 options of gel implants for study patients. Motiva Implant SmoothSilk

Dr Glicksman is an assistant clinical professor, Hackensack Meridian School of Medicine, Nutley, NJ; and is a clinical editor for *Aesthetic Surgery Journal*. Dr Wolfe is a plastic surgeon in private practice in Golden, CO. Dr McGuire is a clinical instructor of surgery, Washington University, St. Louis, MO; and is a clinical editor for *Aesthetic Surgery Journal*.

Corresponding Author:

Dr Caroline Glicksman, 2164 State Highway 35, Building A, Sea Girt, NJ 08750, USA.

E-mail: docmomcag@gmail.com

Round with ProgressiveGel has a slightly firmer gel and a thicker shell than the Motiva Implants SmoothSilk Round Ergonomix ProgressiveGel Ultima, referred to as Round Ergonomix implant (Ergonomix1). The SilkSurface of the Motiva implant is created by 3-dimensional (3D) inverted negative imprinting. A textured mandrel is dipped in silicone during the manufacturing process, transferring a negative image of the 3D mandrel surface to the inside of the shell as the silicone cures. The shell is then inverted before being filled with silicone gel, reorienting the surface architecture to the outside of the implant. In 2024, the International Organization for Standardization (ISO) updated their classification of implant surfaces based on the surface finish technique. The ISO reclassified the SilkSurface as an "intentionally modified surface with a surface texture of 4 microns." In contrast, the macrotextured implants associated with breast implant-associated anaplastic large cell lymphoma (BIA-ALCL) have a surface texture of over 100 microns.⁶ Establishment Labs has developed additional modifications to their implants, found in the Ergonomix2 SmoothSilk implant, currently not available in the United States. Motiva SmoothSilk breast implants received FDA approval for primary augmentation and revisional augmentation in September 2024, becoming the fifth breast implant manufacturer to enter the US market.⁷

METHODS

This prospective, multicenter study included 451 patients enrolled in the primary augmentation cohort and 109 patients in the revisional augmentation cohort at 21 investigator sites. The study included genetic females ages 22 years and over, willing to follow all study requirements. Patients were enrolled for primary breast augmentation to increase breast size and revisional procedures to correct or improve the results of a previous breast augmentation procedure, primary breast reconstruction, or revisional breast reconstruction. Patients were excluded for a history of any breast disease considered to be premalignant in 1 or both breasts, a history of connective tissue or autoimmune disorders or rheumatologic disease, or any medical condition that might result in elevated or significant postoperative complications. Patient demographic data were collected at study enrollment ([Supplemental Table 1](https://doi.org/10.1093/asj/sjaf245), available online at <https://doi.org/10.1093/asj/sjaf245>). Enrollment started in April 2018 and was completed in August 2019. The study design was first posted on clinicaltrials.gov (NCT03579901) in July 2018 and last updated on October 31, 2024. Study investigators were given the option to select from 2 styles of implants; the Motiva SmoothSilk Ergonomix and the SmoothSilk Round. For the clinical trial, both styles were available with an optional passive radio frequency identification (RFID) transponder. The MRI subgroup consists of treated patients, willing to undergo MRI, invited to participate in the MRI cohort. Patients are scheduled for breast MRI at Years 1, 2, 3, 5, 7, and 10. Two hundred and eighteen patients enrolled in the MRI cohort, with 152 completing scheduled MRI at Year 5 (79.2% out of the expected due). Patients are required to obtain the MRI at a local radiological facility specifically trained on the study protocol. All MRIs were performed and evaluated by 2 radiologists who were masked to each other's reading. An independent radiologist reading was employed if there was a disagreement on the MRI, and an explantation was not performed. Patients with and without the RFID transponder were included in the MRI cohort. Clinical data were predominantly collected at in-person visits, although virtual patient monitoring was approved by the FDA in 2020 to facilitate study continuation through the COVID-19 pandemic.

Data Collection and Statistical Analysis

Safety Assessment

Safety assessment was performed by cohort at both patient and implant levels. Kaplan-Meier analysis with 95% confidence intervals of any patient complications included adverse events, severe adverse events, and reoperations. Data collected at each annual visit included annual mammography and ultrasound when indicated, photography, manual measurements, Vectra 3D simulations (Canfield Scientific, Parsippany, NJ), adverse events, and any new-onset connective tissue disease (CTD) signs and symptoms or diagnosis. (Patients with diagnosed CTD at baseline were excluded from the study.) The BREAST-Q (Version 2.0) has been included in data collection since 2021. Additional data were collected on new-onset breast cancer and BIA-ALCL. A Cox regression analysis was performed for complications, reoperations, and adverse events. Explanatory variables, including age, race, smoking status, weight, surgical approach, implant pocket location, incision size, implant volume, irrigation, and type of SmoothSilk device, were previously described in the Motiva 3-year clinical data.⁷

Effective Analysis

Effectiveness out to 5 years was calculated based on a performance goal, using a 5-point Likert scale to rate both physician and patient satisfaction. Quality-of-life questionnaires were completed by patients at Years 1, 2, 3 and 5. The BREAST-Q has been completed at each annual visit since 2021.

RESULTS

Follow-up By Cohort

The augmentation cohorts enrolled 560 genetic females, with 451 primary augmentation patients and 109 revisional augmentation patients. The age range was 22 to 69 in both cohorts with a mean age of 33.5 in the primary augmentation cohort and 44.5 in the revisional augmentation cohort. Patient compliance in the primary augmentation cohort was 88.9% at Year 4 and 88.5% at Year 5, and compliance in the revisional augmentation cohort was 79.2% at Year 4 and 84.8% in Year 5. Overall, the expected patient compliance (560 patients minus patients who removed study devices, exited the study, or died) was 87.8% at 5 years. There were 3 deaths in Year 5; 2 in the primary augmentation cohort and 1 in the revisional cohort. Seven patients were explanted and replaced with nonstudy devices and 3 patients in each cohort exited the study at their own request ([Table 1](#)). The MRI subgroup follow-up included breast MRIs at Years 1, 2, 3, and 5. MRI compliance at Year 5 was 79.2% for primary augmentation patients and 78.8% for revisional augmentation patients. The leading reasons for MRI discontinuation were removal of the primary device (19.8%) and patients unavailable for study, which included medical conditions such as pregnancy (4%) and claustrophobia (2%). Among the MRI cohort, 33 participants received RFID-enabled devices. There were no adverse events reported related to RFID devices or radiologist-impaired detection of silent ruptures ([Table 2](#)).

Complications and Reoperations

A 5-year Kaplan-Meier risk analysis was performed for local breast implant complications and was reported by patient and implant. For the primary augmentation cohort, the overall reported reoperation rate at Year 5 was 8.8%. Breast implant complications included

Table 1. Participant Follow-up Compliance: Primary and Revisional Augmentation Cohorts

	By patient		
Accounting by window	Primary augmentation (n = 451)	Revisional augmentation (n = 109)	Overall augmentation (n = 560)
Week 3-6			
Due ^a	451	109	560
Discontinued			
Deaths	0	0	0
Explanted with non-study replacement	0	0	0
Patient request	0	0	0
Lost to follow-up	0	0	0
Other	0	0	0
Seen	446	109	555
Expected ^b	451	109	560
% compliant out of expected	98.9%	100.0%	99.1%
Year 1			
Due ^a	451	109	560
Discontinued			
Deaths	0	1	1
Explanted with non-study replacement	1	0	1
Patient request	1	1	2
Lost to follow-up	0	0	0
Other	0	0	0
Seen	437	101	538
Expected ^b	450	108	558
% compliant out of expected	97.1%	93.5%	96.4%
Year 2			
Due ^a	451	109	560
Discontinued			
Deaths	1	1	2
Explanted with non-study replacement	1	2	3
Patient request	2	2	4
Lost to follow-up	1	0	1
Other	0	0	0
Seen	428	94	522
Expected ^b	449	106	555
% compliant out of expected	95.3%	88.7%	94.1%

Table 1. Continued

Accounting by window	By patient		
	Primary augmentation (n = 451)	Revisional augmentation (n = 109)	Overall augmentation (n = 560)
Year 3			
Due ^a	451	109	560
Discontinued			
Deaths	1	1	2
Explanted with non-study replacement	1	2	3
Patient request	2	2	4
Lost to follow-up	2	0	2
Other	0	0	0
Seen	415	94	509
Expected ^b	449	106	555
% compliant out of expected	92.4%	88.7%	91.7%
Year 4			
Due ^a	451	109	560
Discontinued			
Deaths	1	1	2
Explanted with non-study replacement	1	2	3
Patient request	3	3	6
Lost to follow-up	3	1	4
Other	0	0	0
Seen	399	84	483
Expected ^b	449	106	555
% compliant out of expected	88.9%	79.2%	87.0%
Year 5			
Due ^a	451	109	560
Discontinued			
Deaths	2	1	3
Explanted with non-study replacement	4	3	7
Patient request	3	3	6
Lost to follow-up	14	5	19
Other	0	0	0
Seen	394	89	483
Expected ^b	445	105	550
% compliant out of expected	88.5%	84.8%	87.8%

^aPatients due for annual study follow-up with the protocol window. ^bExpected patients are the number of patients expected within their study window, minus subjects who were explanted, lost to follow-up, deaths, or discontinued.

Table 2. MRI Follow-up Compliance: Primary and Revisional Augmentation Patients

	By patient		
Accounting by window	Primary augmentation (n = 176)	Revisional augmentation (n = 42)	Overall augmentation (n = 218)
Year 1			
Due	176	42	218
Discontinued			
Claustrophobic	2	1	3
Deaths	0	0	0
Primary devices removed	1	0	1
Unavailable ^a	2	2	4
Patient decision	2	0	2
Seen	160	37	197
Expected ^b	171	39	210
% compliant out of expected	93.6%	94.9%	93.8%
Year 2			
Due	176	42	218
Discontinued			
Claustrophobic	3	2	5
Deaths	0	0	0
Primary devices removed	2	4	6
Unavailable ^a	7	0	7
Patient decision	2	0	2
Seen	149	32	181
Expected ^b	164	36	200
% compliant out of expected	90.9%	88.9%	90.5%
Year 3			
Due	176	42	218
Discontinued			
Claustrophobic	5	2	7
Deaths	0	0	0
Primary devices removed	2	5	7
Unavailable ^a	7	2	9
Patient decision	2	0	2
Seen	142	30	172
Expected ^b	162	33	195
% compliant out of expected	87.7%	90.9%	88.2%

Table 2. Continued

	By patient		
Accounting by window	Primary augmentation (n = 176)	Revisional augmentation (n = 42)	Overall augmentation (n = 218)
Year 5			
Due	176	42	218
Discontinued			
Claustrophobic	3	2	5
Deaths	1	0	1
Primary devices removed	5	6	11
Unavailable ^a	8	1	9
Patient decision	2	1	3
Seen	126	26	152
Expected ^b	159	33	192
% compliant out of expected	79.2%	78.8%	79.2%

^aUnavailable includes patients who were pregnant or who had other temporary medical conditions/circumstances preventing an MRI. These patients will return to the denominator for compliance calculations after resolution of the medical condition/circumstance. ^bExpected indicates the number due minus all discontinuations except patient decision.

Table 3. Kaplan-Meier Risk Analysis at Year 5 for Key Events by Patient

Event	Primary augmentation (n = 451)	Revisional augmentation (n = 109)
Reoperation	38 (8.8%)	37 (36.0%)
Explantation	13 (3.1%)	23 (22.8%)
Explantation with replacement	11 (2.6%)	19 (19.2%)
Explantation without replacement	2 (0.5%)	4 (4.1%)
Number of patients with at least 1 reoperation	38 (8.4%)	37 (33.9%)
Risk of any complication ^a	52 (12.0%)	41 (39.5%)
Breast pain	5 (1.2%)	2 (2.2%)
Animation deformity	1 (0.2%)	0
Asymmetry	1 (0.2%)	4 (3.9%)
Cancer-breast (new)	1 (0.3%)	0
Capsular contracture Grade II with surgical intervention	1 (0.2%)	2 (1.9%)
Capsular contracture III/IV	2 (0.5%)	8 (7.9%)
BIA-ALCL	0	0
Double capsule	0	1 (1.0%)
Infection	4 (0.9%)	1 (0.9%)
Rupture-suspected/confirmed	1 (0.6%)	0

Table 3. Continued

Event	Primary augmentation (n = 451)	Revisional augmentation (n = 109)
Size change	4 (10.0%)	11 (27.5%)
Delayed wound healing	1 (0.2%)	0
Hematoma	3 (0.7%)	2 (1.8%)
Iatrogenic injury to implant	0	1 (1.0%)
Implant extrusion	0	1 (1.0%)
Malposition	17 (3.9%)	6 (6.0%)
Palpability/visibility	2 (0.5%)	0
Breast mass/cyst/lump	3 (0.7%)	2 (2.2%)
Ptosis	1 (0.2%)	6 (6.0%)
Nipple complication	1 (0.2%)	0
Skin rash	1 (0.2%)	0
Seroma (late > 1 year)	0	0
Wrinkling/rippling	2 (0.5%)	0

BIA-ALCL, breast implant-associated anaplastic large cell lymphoma. ^aDoes not include complications assessed as mild severity, except for capsular contracture III/IV, implant extrusion, and suspected or confirmed rupture.

revision for size change (10%), implant malposition (3.9%), capsular contracture Grade III/IV (0.5%), suspected or confirmed rupture (0.6%), glandular ptosis (0.2%), and asymmetry (0.2%). In the revisional augmentation cohort, the 5-year reported reoperation rate for any

Table 4. Risk of Non-local Complications at Each Time Point: Connective Tissue Diseases and Rheumatologic Illness

Complication/time point	Number of events (cumulative)	Kaplan-Meier risk	95% CI
Primary breast augmentation			
Rheumatic disease, any ^a			
Week 6	0 (0)	NA	
Year 1	0 (0)	NA	
Year 2	1 (1)	0.2%	(0.0%, 1.6%)
Year 3	0 (1)	0.2%	(0.0%, 1.6%)
Year 4	0 (1)	0.2%	(0.0%, 1.6%)
Year 5	0 (1)	0.2%	(0.0%, 1.6%)
Hashimoto thyroiditis			
Week 6	0 (0)	NA	
Year 1	0 (0)	NA	
Year 2	1 (1)	0.2%	(0.0%, 1.6%)
Year 3	0 (1)	0.2%	(0.0%, 1.6%)
Year 4	0 (1)	0.2%	(0.0%, 1.6%)
Year 5	0 (1)	0.2%	(0.0%, 1.6%)
Rheumatoid arthritis			
Week 6	0 (0)	NA	
Year 1	0 (0)	NA	
Year 2	0 (0)	NA	
Year 3	0 (0)	NA	
Year 4	0 (0)	NA	
Year 5	0 (0)	NA	
Systemic lupus			
Week 6	0 (0)	NA	
Year 1	0 (0)	NA	
Year 2	0 (0)	NA	
Year 3	0 (0)	NA	
Year 4	0 (0)	NA	
Year 5	0 (0)	NA	
Other rheumatologic			
Week 6	0 (0)	NA	
Year 1	0 (0)	NA	
Year 2	0 (0)	NA	
Year 3	0 (0)	NA	

Table 4. Continued

Complication/time point	Number of events (cumulative)	Kaplan-Meier risk	95% CI
Year 4	0 (0)	NA	
Year 5	0 (0)	NA	
Revisional breast augmentation			
Rheumatic disease, any ^a			
Week 6	0 (0)	NA	
Year 1	0 (0)	NA	
Year 2	0 (0)	NA	
Year 3	0 (0)	NA	
Year 4	0 (0)	NA	
Year 5	0 (0)	NA	
Rheumatoid arthritis			
Week 6	0 (0)	NA	
Year 1	0 (0)	NA	
Year 2	0 (0)	NA	
Year 3	0 (0)	NA	
Year 4	0 (0)	NA	
Year 5	0 (0)	NA	
Systemic lupus			
Week 6	0 (0)	NA	
Year 1	0 (0)	NA	
Year 2	0 (0)	NA	
Year 3	0 (0)	NA	
Year 4	0 (0)	NA	
Year 5	0 (0)	NA	
Other rheumatologic			
Week 6	0 (0)	NA	
Year 1	0 (0)	NA	
Year 2	0 (0)	NA	
Year 3	0 (0)	NA	
Year 4	0 (0)	NA	
Year 5	0 (0)	NA	

CI, confidence interval; NA, not applicable. ^aRheumatologic and connective tissue diseases include rheumatoid arthritis, systemic lupus erythematosus, discoid lupus, scleroderma, vasculitis, dermatomyositis, Raynaud's phenomenon, Sjogren's syndrome, CREST syndrome, carpal tunnel syndrome, multiple sclerosis-like syndrome, multiple myeloma-like syndrome, chronic fatigue syndrome, and fibromyalgia.

Table 5. Satisfaction with Breast Implants on Breast-Q at Year 5: Primary Augmentation

By patient	n	Dissatisfied	Satisfied
Amount of cleavage when wearing a bra	359	5.0%	95.0%
Bra fit	360	5.0%	95.0%
Breast firmness	360	4.2%	95.8%
Breast shape without a bra	359	5.8%	94.2%
Breast size	359	6.4%	93.6%
Breast size matches the rest of the body	360	4.2%	95.8%
Breast feels to touch	358	3.4%	96.6%
How close breasts are when not wearing a bra	358	9.5%	90.5%
How close breasts match with each other	359	7.5%	92.5%
Implant evenness/position relative to each other	359	7.0%	93.0%
Implant position on the chest	358	8.1%	91.9%
Look in the mirror clothed	360	4.4%	95.6%
Look in the mirror unclothed	356	9.8%	90.2%
Natural look of breasts	357	4.5%	95.5%
How naturally breasts sit/hang	359	6.7%	93.3%

reason was 36.0%, and the number of patients who underwent at least 1 reoperation was 33.9%. This included explantation with replacement (19.2%) and explantation without replacement (4.1%). The overall risk of any complication (39.5%) included revision for size change (27.5%), capsular contracture Grade III/IV (7.9%), implant malposition (6.0%), ptosis (6.0%), and asymmetry (3.9%). In the revisional augmentation cohort there was 1 documented iatrogenic rupture of a breast implant (1.0%) and 1 patient with an implant extrusion (1.0%) (Table 3).

Similar to other pivotal studies, patients with a known medical history of connective tissue disease (CTD) and rheumatologic disease, including lupus and scleroderma, were excluded from enrollment in the clinical trial. Patients were asked to document any new diagnosis of CTD or rheumatologic illness at each annual visit. There was 1 patient diagnosed with Hashimoto thyroiditis at Year 2 (0.2%). There were no additional reports in either cohort of new onset of CTD or rheumatologic illness at Years 4 or 5. The FDA has not identified a clear association between CTD and silicone breast implants (Table 4).

BIA-ALCL, BIA-SCC, Breast Cancer

There were no reports of BIA-ALCL or breast implant-associated squamous cell cancer (BIA-SCC) through Year 5. There was 1 (0.3%) new breast cancer diagnosis (invasive ductal carcinoma) reported in the primary augmentation cohort.

Clinical Effectiveness and Patient Satisfaction

Physician and patient satisfaction rates remained high through Year 5. Patient satisfaction data collected using the BREAST-Q

Table 6. Satisfaction With Breast Implants on Breast-Q at Year 5: Revisional Augmentation

By patient	n	Dissatisfied	Satisfied
Amount of cleavage when wearing a bra	76	3.9%	96.1%
Bra fit	75	4.0%	96.0%
Breast firmness	75	4.0%	96.0%
Breast shape without a bra	76	11.8%	88.2%
Breast size	76	6.6%	93.4%
Breast size matches the rest of the body	76	1.3%	98.7%
Breast feels to touch	76	3.9%	96.1%
How close breasts are when not wearing a bra	76	11.8%	88.2%
How close breasts match with each other	76	10.5%	89.5%
Implant evenness/position relative to each other	76	10.5%	89.5%
Implant position on the chest	76	9.2%	90.8%
Look in the mirror clothed	76	3.9%	96.1%
Look in the mirror unclothed	76	11.8%	88.2%
Natural look of breasts	76	7.9%	92.1%
How naturally breasts sit/hang	76	11.8%	88.2%

demonstrated satisfaction rates of 90.2 to 96.6% in the primary augmentation cohort and 88.2% to 98.7% in the revisional augmentation cohort (Tables 5, 6). Patient satisfaction also was measured with a 5-point Likert Scale, and demonstrated high overall satisfaction by patient at 5 years in both the primary augmentation (96.1%) and in revisional augmentation (91.3%) cohorts (Tables 7, 8).

DISCUSSION

This report includes the 5-year data from the US multicenter clinical trial. To encourage a high rate of follow-up compliance, both patients and investigators receive financial incentives at fair market value (FMV). In addition, those patients that complete the full 10-year study will receive a unique additional incentive for their participation. Including financial incentives was not novel in this trial and continues to be an effective mechanism to encourage high patient and physician compliance. The FDA's acceptance of virtual consultations when absolutely necessary, initiated during the COVID pandemic, allowed many patients to remain in the 10-year trial despite missing an in-person consultation. Patients were able to attend a virtual meeting with the investigator and complete all necessary quality-of-life and health questionnaires, returning to in-person visits the following year.

MRI provides valuable data on device integrity not achieved in some earlier clinical trials. The MRI cohort has maintained high patient compliance (79%) at Year 5. Patients were considered unavailable for MRI if they were pregnant or had temporary medical conditions or circumstances preventing MRI. The suspected or confirmed rupture rate of 0.6% in the primary augmentation cohort has remained unchanged from Year 3 to Year 5. There were no ruptures reported in the revisional augmentation cohort through Year 5. The category of iatrogenic implant rupture has been added to the rupture

Table 7. Patient Satisfaction at Year 5: 5-Point Likert Scale

Satisfaction by patient ^a	Augmentation (n = 387)	Revisional augmentation (n = 80)	Overall augmentation (n = 467)
Satisfaction			
Very satisfied	311 (80.4%)	57 (71.3%)	368 (78.8%)
Satisfied	61 (15.8%)	16 (20.0%)	77 (16.5%)
Neither dissatisfied nor satisfied	8 (2.1%)	5 (6.3%)	13 (2.8%)
Dissatisfied	6 (1.6%)	2 (2.5%)	8 (1.7%)
Very dissatisfied	1 (0.3%)	0	1 (0.2%)
Satisfaction summary			
Satisfied	372 (96.1%)	73 (91.3%)	445 (95.3%)
Neither	8 (2.1%)	5 (6.3%)	13 (2.8%)
Dissatisfied	7 (1.8%)	2 (2.5%)	9 (1.9%)
Satisfaction analysis			
Percent satisfied	96.1%	91.3%	95.3%
95% CI	(93.7%, 97.8%)	(82.8%, 96.4%)	(93.0%, 97.0%)

CI, confidence interval. ^aIncludes only patients with non-missing responses for primary implants at this follow-up year (5).

Table 8. Investigator Satisfaction at Year 5: 5-Point Likert Scale

Satisfaction by investigator ^a	Augmentation (n = 387)	Revisional augmentation (n = 80)	Overall augmentation (n = 467)
Satisfaction			
Very satisfied	338 (87.3%)	66 (82.5%)	404 (86.5%)
Satisfied	44 (11.4%)	12 (15.0%)	56 (12.0%)
Neither dissatisfied nor satisfied	3 (0.8%)	0	3 (0.6%)
Dissatisfied	0	2 (2.5%)	2 (0.4%)
Very dissatisfied	2 (0.5%)	0	2 (0.4%)
Satisfaction summary			
Satisfied	382 (98.7%)	78 (97.5%)	460 (98.5%)
Neither	3 (0.8%)	0	3 (0.6%)
Dissatisfied	2 (0.5%)	2 (2.5%)	4 (0.9%)
Satisfaction analysis			
Percent satisfied	98.7%	97.5%	98.5%
95% CI	(97.0%, 99.6%)	(91.3%, 99.7%)	(96.9%, 99.4%)

CI, confidence interval. ^aIncludes only patients with non-missing investigator responses for primary implants at this follow-up year (5).

data, documented only when the investigator surgeon confirms that there was inadvertent damage to the device during a surgical procedure.

There were 2 patients who underwent a reoperation for capsular contracture (Grade III/IV) in the primary augmentation cohort by Year 5. The low capsular contracture rates observed in this pivotal study validate the resurgence in research focused on the influence

of surface characteristics on the local inflammatory process. The early cellular response to the presence of an implant and potential bacterial attachment and biofilm formation is affected by a combination of surface technology and regimented bacterial mitigation strategies.⁸

There were 10 reoperations in the revisional augmentation cohort for recurrent capsular contracture. In this cohort, 15 patients were

enrolled due to implant rupture and 20 patients for Grade III/IV capsular contracture. Of the patients enrolled with previous implant rupture, 4 of 15 (27.0%) developed a subsequent capsular contracture requiring a revision. Of the patients enrolled for capsular contracture, 4 of the 20 (20%) developed a recurrent capsular contracture requiring surgical intervention. The recurrent capsular contractures occurred at 5 investigator sites. Each of the investigators reported that they performed a total capsulectomy at the time of initial Motiva placement.

Recurrent capsular contracture rates in patients treated for established capsular contracture have been documented in previous pivotal studies to range from 20% to 21%.^{9,10} Capsular contracture (Grade III/IV) after revisional augmentation surgery is documented to have a high rate of recurrence with capsulectomy alone (27.2%).¹¹ Recently published studies described revisional protocols that include capsulectomy (total or partial), site change, implant exchange, antibiotics and povidone solutions, and specify the use of acellular dermal matrix (ADM) or scaffold.^{12,13} The Motiva pivotal trial protocol did not permit scaffold or ADM in the management of recurrent capsular contracture.

Data were collected after implantation to document any new onset of CTD or autoimmune disorders. Over 5 years, 1 patient reported a diagnosis of Hashimoto's thyroiditis. There have been no other new diagnoses of lupus, rheumatoid arthritis, scleroderma or other rheumatologic illness in either cohort. Patients will continue to be monitored for CTD and rheumatologic disorders for the full 10 years.

Size change remains the leading reason for reoperation in both the primary augmentation (10%) and revisional augmentation (27%) cohorts at 5 years. Revision for size change remains a subjective decision between the investigator surgeon and the patient and is not considered device related.

The limitations of this study include the fact that the investigators were experienced plastic surgeons with previous involvement in clinical trials and that they selected patients who were suitable candidates for surgery and willing to participate in a 10-year study. Furthermore, because 94% of the participants received implants placed in a dual-plane pocket, the reported 5-year results may not be applicable to patients undergoing subfascial or subglandular implant placement.

CONCLUSIONS

The Motiva clinical trial began patient enrollment in 2018, with some sites now in their seventh year of patient follow-up. These data represent the 5-year submission to the FDA in the primary augmentation and revisional augmentation cohorts. The Motiva clinical trial introduced US surgeons to several new modifications, including the Motiva SmoothSilk surface, which was intentionally designed to lower rupture and capsular contracture rates. Clinical trial evidence continues to demonstrate very low rupture rates and low capsular contracture rates, with high levels of satisfaction reported by patients and surgeons alike.

Supplemental Material

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Disclosures

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